ISE549 – Management of Innovation & Technology

Week 9: Managing New Product Development (NPD)

Complete English Transcript of the Lecture
(Cleaned & structured from Audio Part 1 / Part 2 / Part 3 + Lecture Slides)

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1. Agenda & Introduction to NPD

Today's agenda: (1) Introduction to Managing New Product Development (NPD); (2) Theory in action / cases; (3) NPD process stages; (4) Tools for improving NPD; (5) Metrics; (6) Best practices and summary.

NPD is a critical competitive advantage for brand owners and industry players. Innovation is not limited to the product itself – it also covers technology, process, product timing and customer perception. NPD requires a well-structured, radical cross-functional approach that extends beyond internal departments to the whole supply chain and business partners, in order to transform raw ideas into market-ready offerings. "Market-ready" does not mean the product has already been launched; it means the offering is ready for market-acceptance testing.

2. The NPD Process (Waterfall / Stage-Gate View)

The classic NPD process can be viewed as a sequence of stages (often presented as a waterfall or stage-gate model):

Stage 1 – Idea Generation: Collect as many ideas as possible (e.g. 50 ideas from 50 students). Ideas are triggered mainly by two drivers: (a) the need to solve an existing problem, and (b) market demand / opportunity.

Stage 2 – Idea Screening: Filter the collected ideas against criteria such as technical feasibility, business-strategy alignment and resource availability (especially rare or specialised resources).

Stage 3 – Concept Development & Testing: Develop early-stage prototypes / samples / MVPs (Minimum Viable Products). An MVP is the simplest version that still delivers the core value (e.g. "something that moves on wheels"). Test the concept and prototype with target users.

Stage 4 – Business Analysis: Evaluate financial viability (cost vs expected market price), sales projections, cost structure and required ROI. Many large organisations have formal corporate ROI hurdles.

Stage 5 – Product Development: Full engineering, assembly, iterative design and testing by cross-functional teams. Deliverable is a real product, not just a concept.

Stage 6 – Market Testing / Pilot (Soft) Launch: Limited pilot launch to pilot users and pilot markets (typically innovators and early adopters on the diffusion curve). Gather feedback and make modifications before full scale-up.

Stage 7 – Commercialisation / Full Launch: Full-scale manufacturing, marketing campaigns, digital channels, supply-chain & logistics, sales & distribution, road-shows, demonstrations and education. Create awareness so that customers know where and how to buy. For high-tech and luxury products, commercialisation cost often exceeds pure production cost because the firm must "tell and sell the whole story".

These stages can be grouped into broader phases: Filter (idea generation + screening) → Plan (concept + business analysis) → Build (product development + market testing) → Launch (commercialisation).

3. The 3I Theory: Idea → Invention → Innovation

• **Idea**: Generation and screening of ideas and thoughts.
• **Invention**: Turning the idea into something concrete (new technology, prototype). At this stage the firm must protect its intellectual property (patent, copyright, trademark).
• **Innovation**: Commercialising the invention into a new product that is launched into the market. Only after successful commercialisation does the invention become a true business innovation.

In an open-innovation environment, ideas and technologies can also flow into the organisation from outside. Technology can be licensed out (licensor receives royalties) or spun out. A familiar example of an open technology platform is the mobile OS (Android / iOS) that allows third-party developers to create and distribute apps.

4. Idea Generation Example – The Selfie Stick

Problem identification: limited arm length prevents taking a long-shot group photo with a smartphone. Idea: extend the arm – a selfie stick. Further ideas quickly appear: remote shutter control (because the button is now too far), stand / tripod base, adjustable height, Bluetooth, lighting, stabilisation, etc. Ideas therefore cover functional features, design form, and the enabling technology. This illustrates that idea generation starts from a real problem or market need and then expands into function, design and technology layers.

5. Idea Screening & Prioritisation Tools

Ideas can be funnelled from many black dots (raw insights) through evaluation layers down to a small set of prioritised green ideas. Customer signals, peer evaluation and predictive scoring help amplify the most promising insights.

RWW screen – Real, Win, Worth it:
• **Real**: Is there a real market / real customer need?
• **Win**: Can we win against competitors and build competitive advantage?
• **Worth it**: Is it profitable and is the risk acceptable (calculated / anticipated risk)?

6. Four Key Tools for Improving the NPD Process

6.1 QFD – Quality Function Deployment (House of Quality)

QFD improves communication and coordination among Marketing, Engineering and Manufacturing. Marketing collects customer requirements; Engineering translates them into design / engineering attributes; Manufacturing turns the design into a producible product. The House of Quality matrix shows the relationships (strong / medium / weak) between customer requirements and engineering attributes, customer importance weights, competitive benchmarking (our product vs competitors A and B), and design targets. Strong positive relationships are typically scored 9. The matrix helps prioritise which engineering attributes will most satisfy important customer needs and where the new design must outperform competitors.

6.2 DFM – Design for Manufacturing

DFM applies design rules that reduce cost and development time while improving quality and speed-to-market. Typical rules: minimise the number of parts; use common / standard parts across a product family; simplify assembly; reduce material handling and inventory; enable automation of repetitive, boring or dangerous tasks; eliminate adjustments and assembly errors; design for environmental compliance; and simplify or eliminate non-value-adding process steps. Standard parts → standard processes → easier automation and lower total production cost (materials, labour and machine).

6.3 FMEA – Failure Mode and Effects Analysis

FMEA systematically identifies potential failure modes before launch and ranks them by Severity (impact), Occurrence / Likelihood (frequency) and Detection / Controllability. A composite risk score prioritises which failures must be prevented or mitigated. Performing FMEA during testing / pilot stages allows failures to be discovered and eliminated early. Without FMEA, failures surface only after full market launch, producing a sharp peak in failure cases and high recovery cost.

6.4 CAD / CAM – Computer-Aided Design & Manufacturing

CAD enables rapid 2D-to-3D design, measurement and inexpensive virtual prototyping. CAM translates the digital model into manufacturing instructions, supporting faster changeovers and more product variation at reasonable cost (especially when modular / standard components are used). 3D printing is a common CAM realisation: the model is sliced into layers and material is deposited layer by layer. Popular tools include AutoCAD and Autodesk. Example: aircraft-engine concept → CAD 3D model → sliced layers → additive manufacturing → finished prototype.

7. Measuring NPD Performance

Measuring NPD helps management: set and track innovation-strategy goals; benchmark performance (against industry leaders, direct competitors and own history); optimise resource allocation and employee incentives; and refine future innovation strategy.

Process metrics (Speed-to-Market focus): average cycle time; variation of cycle time by project type (breakthrough vs incremental); percentage of projects that met deadline; percentage that stayed within budget; percentage that resulted in a completed product.

These three dimensions – Time, Cost and Quality – form the classic **triple constraint** of project management. An NPD project manager must control all three tightly.

Success-in-Market / Return-on-Innovation metrics: percentage of products that achieve their sales targets; percentage of revenue generated by products launched in the last 1–5 years; ratio of successful projects to total projects in the portfolio; overall return on innovation (profit from new products versus total expenditure across R&D, engineering, production, commercialisation and marketing).

Google's post-mortem practice is cited as a best-practice example: after every major project the team spends 3–6 months producing a frank 10-to-100-page report that documents what worked, what did not, and concrete areas for improvement. The report is used for internal benchmarking and learning.

8. Best Practices for Effective NPD Management

- **Customer focus** – start from real market demand, customer pain points and problems; idea generation follows problem identification.
- **Cross-functional collaboration** – involve product managers, designers, engineers, quality, marketing and sales (and external partners) from day one.
- **Agile / flexible loops & phase-gate governance** – respond quickly to change; use stage gates for disciplined go/kill decisions while keeping the process interactive.
- **Data-driven decision-making** – track key metrics (time-to-market, yield, cost variance, quality) and support both sequential and parallel decision styles as appropriate.

• **Structured methodologies** – leverage frameworks such as Product Lifecycle Management (PLM) software to centralise data, and combine internal capabilities with external resources.

9. Closing Reflections & Mind-set

Innovation often begins with a problem (e.g. environmental pressure → electric vehicles; limited arm length → selfie stick). The invention of the light bulb did not come from improving the candle; it came from a different way of thinking about illumination. Apple's "Think Different" message and the psychological capital model (Hope, Efficacy, Resilience, Optimism – HERO / PsyCap) are offered as reminders to stay positive, learn from failure ("FAIL = First Attempt In Learning", "END = Effort Never Dies", "NO = Next Opportunity"), and tolerate both positive and critical feedback (Jeff Bezos). A short self-study case on the hard-disk-drive industry and further AI-generated selfie-stick concept ideas are provided for optional exploration.

10. Exam-Friendly Key Takeaways

• NPD is a structured, cross-functional process that turns ideas into market-ready (and ultimately commercialised) offerings.

• Classic stages: Idea Generation → Screening → Concept & Testing → Business Analysis → Product Development → Pilot/Market Test → Commercialisation.

• 3I path: Idea → Invention (protect IP) → Innovation (commercialisation).

• Four core tools: QFD (House of Quality), DFM, FMEA, CAD/CAM.

• Measure both process (time, cost, quality – the triple constraint) and market success / return on innovation.

• Best practices: customer focus, cross-functional teams, agile + stage-gate, data-driven decisions, structured PLM support.

• Start from real problems or market demand; protect IP before full commercialisation; learn systematically after every project.

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Sources: ISE549 W9 GN.pdf + ISE549_W09_Part1/2/3.m4a. Prepared for study purposes. Content follows the original lecture by Prof. Andy C.Y. Sun.